MACHINE LEARNING ENGINEER ASSESSMENT

Freight rate prediction

Estimating the total posted price of a shipment from its route, equipment, weight, pickup date and market conditions.

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02 RESULTS

Forecast and historical accuracy

The model predicts the posted rate in US dollars for one load. Historical evaluation covers April through October 2025. The final batch covers November and December.

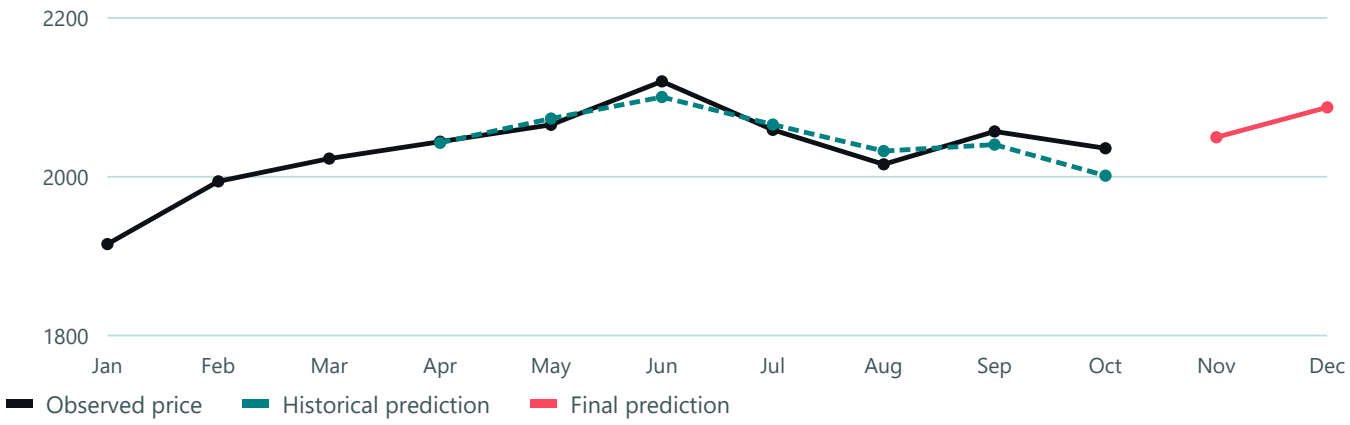
$89.35
Mean of seven monthly MAEs

$25.11
Median absolute error

1.39%
Prices outside 0.5x to 2x

Monthly posted rates

Median total rate per shipment, USD. Each historical prediction uses a model fitted before its forecast window began.



Month-to-month changes reflect both the market and the mix of shipments. November and December show predictions for the 12,000 loads with undisclosed prices.

Typical accuracy and the cost of rare misses

Across seven validation months, the median absolute error stayed between $22.55 and $28.28. Monthly mean absolute error (MAE) ranged from $77.10 to $106.36. The largest 338 errors account for 55.46% of the total absolute error.

For 468 of 33,709 loads, the posted price was below half or above twice the forecast. Most of their recorded shipment characteristics look ordinary. A later-period detection test found little ability to identify these cases before the price was known.

03 DATA

What the shipment data reveal

The 48,000 labeled loads from January through October had a median posted rate of $2,030.76 and a median distance of 953 miles. The final batch contains 12,000 additional loads.

Route and distance

Distances range from 70 to 3,439.8 miles. In a controlled experiment with the initial two-stage model, removing city effects raised mean monthly MAE from $92.08 to $108.13. A model using distance alone, with a flexible nonlinear curve, scored $196.76.

Calendar and market conditions

Differences between dates account for about 98% of the historical variation in the market index. After adjusting for the index, estimates using ten days on either side of a quarter boundary showed drops of 8.5 cents per mile around April 1 and 9.4 cents around July 1. The model allows rates to vary with progress through the calendar quarter.

An unstable quote signal

The correlation between quote_signal and the actual price per mile was +0.49 in March, -0.47 in April and near zero in August. Its relationship to price changed direction within the training period. The selected model uses the market index and omits the preliminary quote.

New cities in the final batch

Training covered 64 cities. Eight new cities appear in November and December. Of the 1,461 loads on previously unseen directed routes, 1,447 involve a new city. Historical validation tested new combinations of familiar cities. Performance on newly introduced cities remains unmeasured.

> Coordinates support straight-line distance and direction features. Duplicate city names and displaced coordinates complicate state assignment. Adding inferred states or regions yielded $89.41 to $89.48 MAE, versus $89.35 for the selected model.

04 DATA QUALITY

Preparing the data

Load identifiers are unique within each batch and all 48,000 training prices are present. Each historical training window supplies its own medians and feature bounds.

Input check	Jan-Oct: 48,000 loads	Nov-Dec: 12,000 loads
Missing weight	300 (0.63%)	165 (1.38%)
Missing market index	374 (0.78%)	249 (2.08%)
Negative weight	292 (0.61%)	145 (1.21%)
Missing city, coordinate, equipment, distance or date	0	0

Weight and equipment

A negative weight contributes its magnitude and a separate indicator for the original sign. A missing weight contributes the training median of absolute weight and a missing-value indicator. The final training median was 31,496 lb. Computing a separate median for each equipment type changed historical MAE by less than one cent in a controlled comparison.

Market index

During training of the two-stage model, missing values use the mean observed index on that date, with the overall training median as a fallback. At prediction time, a missing index uses the fitted calendar forecast of the market. An indicator records whether the row supplied its own index.

Extreme features and prices

Additional nonlinear weight and distance features are bounded at the 0.5th and 99.5th training percentiles. The model retains the original target price for every shipment. Huber loss limits the influence of large errors during training, with a $60 transition point. Evaluation counts each absolute dollar error in full.

Page 7 examines the effect of removing specific extreme prices after evaluation. That calculation uses the observed price, which is unavailable when quoting a new load.

05 METHOD

Model and validation

The two-stage regression separates the price contribution of individual shipment characteristics from the market level on the pickup date.

1. Estimate shipment effects within each date

The first stage fits distance, equipment, weight, route geometry and directional city effects alongside a separate effect for each training date. This compares loads shipped on the same day and allows both route costs and the daily market level to vary.

2. Forecast the daily rate level

The second stage projects the date effect using time trend, weekday, quarter progress and market conditions. A simpler trend model handles short histories. The selected model adds proximity to observed US federal holidays and prior forecast errors on the same directed route, with an equipment-specific adjustment. Sparse routes receive smaller corrections.

Configuration on identical evaluation loads	Mean monthly MAE
Initial two-stage regression	$92.08
Add geometry, direction and cargo features	$91.39
Add prior route errors to that specification	$89.79
Add holiday proximity to that specification	$90.88
Use both route errors and holiday proximity	$89.35

Expanding-window cross-validation

Each fit predicts two subsequent months. Training through March predicts April-May, through May predicts June-July, through June predicts July-August, and through August predicts September-October. For July, the later forecast origin takes precedence. Each of the 33,709 evaluation loads counts once.

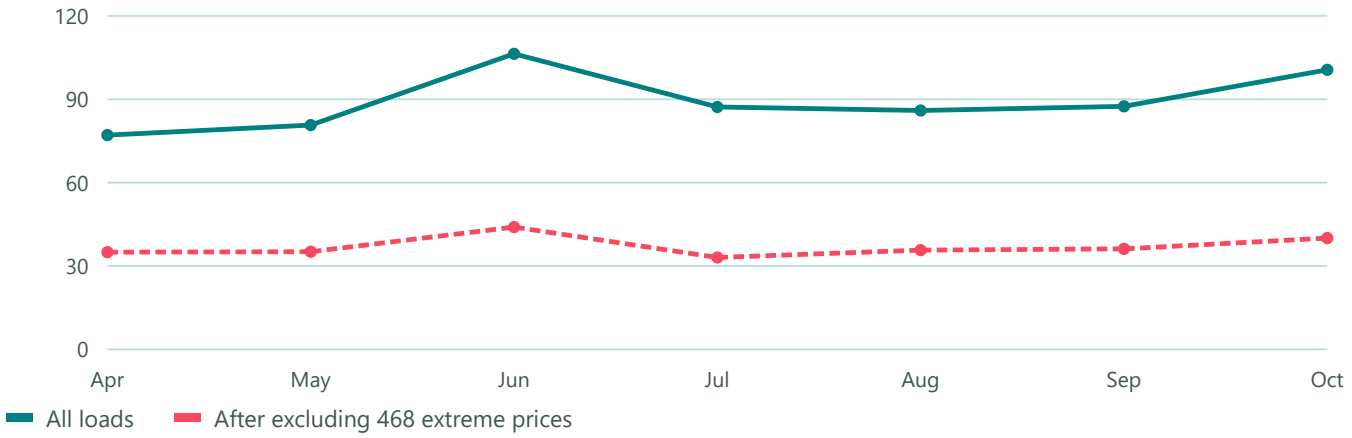
These windows informed feature selection and yielded $89.35 mean monthly MAE. The final model was then fitted on all 48,000 labeled rows and applied to the 12,000 November-December loads. Prices for those loads are undisclosed.

On the same development periods, direct RealMLP scored $107.37. Residual GRU and Transformer models averaged $93.13 and $92.04 respectively across three seeds.

06 VALIDATION

Accuracy by month

The primary score gives equal weight to each of seven calendar months. Within a month, mean absolute error gives every shipment equal weight and measures the dollar difference between prediction and posted price.



Month	Loads	MAE, all	Median abs. error	Subset MAE*
Apr	4,819	$77.10	$24.73	$34.97
May	4,913	$80.71	$24.55	$35.15
Jun	4,783	$106.36	$28.28	$44.01
Jul	4,912	$87.23	$22.55	$33.07
Aug	4,759	$85.96	$24.76	$35.72
Sep	4,670	$87.46	$23.74	$36.16
Oct	4,853	$100.60	$28.01	$40.05
Overall	33,709	$89.35	$25.11	$37.02

The overall row reports the pooled median absolute error and the unweighted means of the monthly MAEs.

> June and October had the largest MAEs. Even in those months, half of predictions were within $29 of the posted price. Adding route-error and holiday features reduced MAE in six of seven months. The largest gain was $7.06 in August. May became $0.88 worse.

* The diagnostic subset excludes loads selected using their observed prices. Page 7 gives the selection rule and its effect.

07 PRICE DIAGNOSTICS

The price extremes

Dividing each recorded price by its historical prediction gives a ratio of 1 for an exact match. The 33,709 observations fall into three visibly separated groups.



Price group	Loads	Actual / predicted	Share
Low	232	0.173 to 0.445	0.69%
Central	33,241	0.897 to 1.100	98.61%
High	236	2.226 to 5.368	0.70%

Why 1% of loads accounts for 55% of error

The 338 largest errors average $4,941 per load, versus $40.19 across the remaining 33,371 loads. Together they contribute 55.46% of total absolute error. All 338 belong to the extreme groups. A further 130 loads have a large relative deviation but a smaller dollar error.

> With identical predictions, mean monthly MAE is $89.35 across all loads, $40.21 after excluding the 338 largest dollar errors and $37.02 after excluding all 468 extreme ratios.

The gap between the groups is consistent with a scale change in some target prices. The available fields cannot establish the cause. Selecting rows by their realized errors makes these subset scores retrospective.

08 DETECTION

Identifying an extreme price before pickup

Of the 338 largest errors, 334 occurred on routes seen in the respective training history. Extreme dollar errors affected 0.91% to 1.06% of loads within each equipment class.

Characteristic	Largest 338 errors	Other 33,371 loads
Median distance	1,227 mi	950 mi
Median absolute weight	31,261 lb	31,544 lb
Missing weight	0.59%	0.58%
Negative weight	1.78% (6 loads)	0.55%
Median market index	1.113	1.135

Selecting on dollar error favors long, expensive trips. Negative weight was more frequent among large misses, but it occurred in just six of the 338 cases.

One lane, comparable prior loads

> Phoenix to Syracuse, Dry Van, 2,811 miles. Five earlier loads with the same direction and equipment and similar mileage were priced at $4,736 to $5,078. The forecast was $5,058 and the posted price was $24,295. Deviations also went the other way. A Phoenix to Philadelphia load was priced at $1,048 against a $5,568 forecast.

A test using only information available before pricing

Two classifiers trained on April-August loads ranked 9,523 September-October loads by risk of an extreme price ratio. Inputs included cities, coordinates, equipment, weight, dates, market fields and prior route counts.

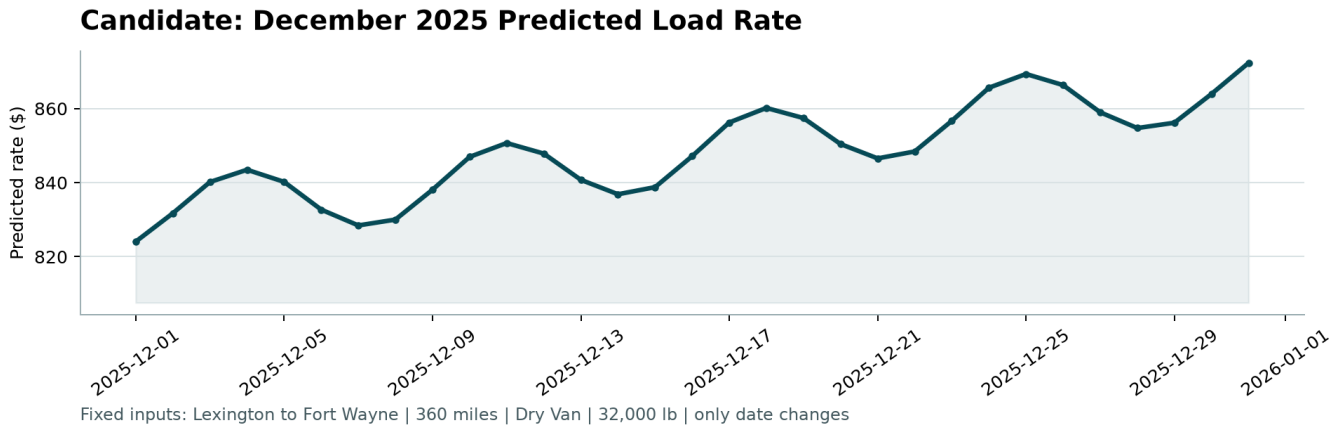
Logistic regression reached ROC AUC 0.510 and gradient boosting reached 0.511, close to the random-ranking value of 0.5. Of the 96 highest-risk loads flagged by boosting, four belonged to the 144 actual extreme cases. These results do not support a reliable correction rule for individual quotes.

09 FINAL PREDICTIONS

The fixed December forecast

The required chart plots one predicted total load price for each day from December 1 to December 31, 2025, holding all shipment characteristics fixed.

> Lexington to Fort Wayne | 360 miles | Dry Van | 32,000 lb



Output of the supplied score.py. Predictions for this fixed shipment range from $824.08 to $872.32.

Two prediction inputs

The main model uses the richer shipment fields provided with all 12,000 November-December loads. The 31-row chart has only pickup, delivery, distance, equipment, weight and date. A separate model fitted for this six-field input estimates market conditions from training history.

The six-field model scored $106.19 mean monthly MAE on historical data. The supplied scorer validates both submission files and draws the chart. Spotter evaluates final predictive accuracy against the undisclosed November-December prices.